

No. of Functional Features Included in the Solution

Date	03 July 2024
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	HDI Prediction	Engine Accepts Life Expectancy, Expected/Mean Years of Schooling and GNI per Capita, and computes the Life Expectancy Index, Education Index, Income Index and final HDI score via the /api/predict endpoint.	Backend – Flask API / predictor.py	Done	1.0
2	Multi-Model	Lets the user switch between the Official Mathematical (geometric-mean) model and four trained ML models – Linear Regression, Random Forest, Gradient Boosting and Neural Network – for HDI inference.	Selection Backend – trainer.py / predictor.py, Frontend – Predict.tsx	Done	0.75
3	Country Preset Auto-Fill	Selecting a country from a searchable dropdown automatically populates the four indicator fields with that country's known socioeconomic values.	Frontend – Predict.tsx, data/countries.ts	Done	0.5
4	Interactive Analytics Dashboard	Displays average/highest/lowest HDI and total inferences as stat cards, plus an HDI trend area chart, a Longevity/Education/Income radar chart, a GNI-vs-Life-Expectancy scatter chart and an HDI-category bar chart, all backed by /api/analytics.	Backend – app.py, Frontend – Dashboard.tsx	Done	0.75

5	Prediction History Log	Stores every prediction the user runs (locally in app context) and lists it in a searchable, filterable (by category/model) and sortable paginated table.	Frontend – History.tsx, useAppContext.tsx	Done	0.5
6	CSV Export of History	Allows the user to export their full prediction history to a CSV file for offline analysis, and to clear all saved logs.	Frontend – History.tsx	Done	0.25
7	Model Information / Transparency Page	Shows the active model's algorithm name, dataset, R2/MAE/RMSE metrics, feature list, the training pipeline steps (data ingestion, preprocessing, model fit, API inference) and version logs.	Backend – app.py (/api/model-info), Frontend – ModelInfo.tsx	Done	0.5
8	System Health & Country Data API.	Provides a lightweight /api/health endpoint for backend status/monitoring and an /api/countries endpoint that serves the sorted list of countries from the cleaned dataset to the frontend, with graceful fallback to local presets if the backend is offline	Backend – app.py, Frontend – api/health.ts, api/countries.ts	Done	0.25

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Interactive Analytics Dashboard, Prediction History Log, Country Preset Auto-Fill, Model Info Page
2	Backend / Logic	3	HDI Prediction Engine, Multi-Model Selection, HDI category classification logic

3	Database / Storage	2	Kaggle HDI dataset loading & preprocessing, trained model (.pkl) storage
4	API / Integration	5	/api/predict, /api/analytics, /api/countries, /api/model-info, /api/health
5	Security / Authentication	1	CORS-restricted API access (no user login required; single-user analytical tool)